

Mohammad Atif Quamar

MSc in Machine Learning at MBZUAI

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ABOUT

Graduate Researcher in Machine Learning at [MBZUAI](#). My research sits at the intersection of foundation model training and mechanistic interpretability, focusing on pretraining and post-training dynamics, machine reasoning, and multi-agent systems. I am particularly interested in how reasoning and internal mechanisms emerge during training, and how understanding them can help us build models that are safer, more robust, and better aligned.

EDUCATION



M.Sc. in Machine Learning

2026 – 2028

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi, UAE

- Relevant Coursework: Machine Learning, Mathematical Foundations of Machine Learning, Principles of Applied Statistics, Deep Learning, Machine Learning with Python



B.Tech. in Computer Science and Biosciences

2020 – 2024

Indraprastha Institute of Information Technology (IIIT), Delhi, India

- Relevant Coursework: Machine Learning, Computer Vision, Natural Language Processing, Artificial Intelligence, Linear Algebra, Multivariate Calculus, Differential Equations

PUBLICATIONS

* Equal contribution

- [C1] **Reliable Chain-of-Thought via Prefix Consistency**
Naoto Iwase, Yuki Ichihara, Mohammad Atif Quamar and Junpei Komiyama
NeurIPS 2026 - Main Track [\[X arXiv\]](#) [\[Globe Page\]](#) [\[Code\]](#)
- [P1] **Self-Adapting Group of Experts for Multi-Agent Reasoning**
Mohammad Atif Quamar, Nurbek Tastan, Karthik Nandakumar, Junpei Komiyama
arXiv, 2026. Under Review. [\[X arXiv\]](#) [\[Globe Page\]](#) [\[Code\]](#)
- [P2] **Adaptive Blockwise Search: Inference-Time Alignment for Large Language Models**
Mohammad Atif Quamar, Mohammad Areeb*, Nishant Sharma, Ananth Shreekumar, Jonathan Rosenthal, Muslum Ozgur Ozmen, Mikhail Kuznetsov and Z. Berkay Celik*
arXiv, 2025. Under Review. [\[X arXiv\]](#) [\[Code\]](#)
- [P3] **Privileged Solutions or Context-Induced Teacher Behaviour? Dissecting On-Policy Self Distillation**
Yuki Ichihara, Naoto Iwase, Mohammad Atif Quamar and Junpei Komiyama
arXiv, 2026. Under Review. [\[X arXiv\]](#) [\[Code\]](#)
- [P4] **Prefix-Denoising Consistency for Diffusion Language Models**
Yuki Ichihara, Naoto Iwase, Mohammad Atif Quamar, Junpei Komiyama
arXiv, 2026. Under Review. [\[X arXiv\]](#) [\[Code\]](#)
- [P5] **Learning Modal-Mixed Chain-of-Thought Reasoning with Latent Embeddings**
Yifei Shao, Kun Zhou, Ziming Xu, Mohammad Atif Quamar, Shibo Hao, Zhen Wang, Zhiting Hu and Biwei Huang
arXiv, 2026. Under Review. [\[X arXiv\]](#)
- [W1] **STARS: Synchronous Token Alignment for Robust Supervision in LLMs**
Mohammad Atif Quamar, Mohammad Areeb*, Mikhail Kuznetsov, Muslum Ozgur Ozmen, and Z. Berkay Celik*
ICML 2026 - Structured Probabilistic Inference & Generative Modeling Workshop [\[OpenReview\]](#) [\[Code\]](#)
- [W2] **Logit-Entropy Adaptive Stopping Heuristic for Efficient Chain-of-Thought Reasoning**
Mohammad Atif Quamar and Mohammad Areeb
NeurIPS 2025 - Efficient Reasoning Workshop. [\[X arXiv\]](#) [\[OpenReview\]](#)
- [W3] **Decoding Histone Modification Signatures of Non-Coding RNAs via FMs**
Nishant Sharma, Mohammad Atif Quamar and Pengtao Xie
NeurIPS 2025 - Workshop on Multi-modal FMs and LLMs for Life Sciences [\[OpenReview\]](#)

RESEARCH EXPERIENCE



Visiting Researcher

MBZUAI, Abu Dhabi, UAE

Feb 2026 – Jul 2026

Advisor: [Dr. Junpei Komiyama](#)

- Working on methods to improve reasoning in language models through reinforcement learning, self-consistency, and test-time inference strategies, to enhance mathematical reasoning and coding performance while reducing unreliable reasoning paths. Publication - [\[C1\]](#).



Research Intern

University of Virginia, Virginia, United States

Sep 2025 – Feb 2026

Advisor: [Dr. Ferdinando Fioretto](#)

- Developed a drift-resilient memory framework for code-execution agents using KL-constrained adapter updates, mitigating embedding distribution shifts during online learning to reduce unsafe code generation without compromising task success rates.



Research Intern

UC San Diego, California, United States

Aug 2025 – Sep 2025

Advisor: [Dr. Biwei Huang](#)

- Proposed a multimodal Chain-of-Thought framework that interleaves text with latent visual representations in VLMs, with a two-stage SFT + RL training setup for modality switching, achieving strong gains across 11 multimodal reasoning benchmarks. Publication - [\[P5\]](#).



Research Intern

Purdue University, Indiana, United States

Mar 2025 – Jul 2025

Advisor: [Dr. Z. Berkay Celik](#)

- Studied Bayesian sampling and inference-time alignment methods for language models, improving response quality and steering outputs toward harmlessness and positive sentiment. Publications - [\[P2\]](#) and [\[W1\]](#).



Research Intern

UC San Diego, California, United States

Jan 2025 – Feb 2025

Advisor: [Dr. Pengtao Xie](#)

- Built a deep learning model predicting 50 histone modification–ncRNA interactions from transcript and genomic context. Publication - [\[W3\]](#).

INDUSTRY EXPERIENCE

Co-Founder

Insituate, New Delhi, India

Sep 2023 – Jan 2025

- Founded Insituate and grew it to \$250K ARR, building on-premise AI agents using open-source LLMs and RAG systems for enterprise use cases; deployments included Singapore Judiciary, Indian High Courts, Mizuho Bank, and PNC Bank. Built RAG systems for financial documents and legal tools that extract insights and spot defects in case files for High Courts in India and Singapore.

AWARDS & ACHIEVEMENTS

- 2026** **Travel Grant.** Received Travel Grant to attend ELLIS Summer School in TU Munich, Germany.
- 2026** **MBZUAI Scholarship.** 100% tuition scholarship for M.Sc. in ML; \$100K scholarship package.
- 2024** **NASSCOM GenAI Cohort.** Selected among the top 15 AI startups in India.
- 2024** **NVIDIA Inception.** Secured \$150K worth of AWS credits.
- 2024** **Microsoft Founder's Hub.** Received \$150K worth of Azure credits.
- 2020** **JEE Main.** Ranked in the top 3% among 1.1 million candidates.
- Present** **Competitive Programming.** 1800+ rated on CodeChef and LeetCode; world top 3%.

ACADEMIC SERVICE

- 2025** **Program Committee.** AAAI AI Alignment Track.
- 2025** **Program Committee.** Efficient Reasoning Workshop at NeurIPS.
- 2024** **Teaching Assistant.** ECO332: Valuation and Portfolio Management at IIITD.
- 2023** **Teaching Assistant.** DES102: Introduction to HCI at IIITD.

REFERENCES

Junpei Komiyama

Assistant Professor at MBZUAI

M.Sc. Supervisor

✉ junpei.komiyama@mbzuai.ac.ae